Research Assignment 2 – The Language of business Analytics

Section 01 – Foundation Concepts of Analytics

01 (i) <u>BUSINESS ANALYTICS</u>
Is the use of data, information technology, mathematical methods to help businesses gain insight on their operations and product performance thereby making better face-based decisions and solving problems

(ii) Real world example – Shoprite finds out some stock don't sell fast. Business Anlyst – Reviews patterns, and number of expired stock per specific period. Then suggests discounts and promotions. Product sells well.

iii) (Longbing Cao, 2016)

02 <u>Data Analytics</u>
Is the process of cleaning, collecting and analysing data through the use of tools such as Excel, SQL, Python and presenting them on dashboards and report, to find patterns ~~and~~ to be used to take actionable insights.

(ii) Example – In transportation and traffic engineering traffic is counted, and sougnted so as to know which roads to expand and where to construct and install traffic lights.

iii) (Longbing Cao, 2016)

03 <u>Business Intelligence (BI)</u>
Usind data, reports and dashboards to help businesses make better, smart and informed decisions.

(ii) Example - Shoprite collects sales data from all its SA stores. Using Power BI, analystis create dashboards showing, low stock items, revenue, top sells. Then see which product needs restocking and which one to stock less.

iii) Thomas H. Davenport, 2006

4 <u>Data-Driven Decision Making (DDDM)</u>

Using data, analystist reports and business history/patterns to make fece-based business decision rather than doing guesswork.

iii) Example - A certain wig is to selling on an online store:
~~tot~~ without data - owner guesses and removes the product.
With DDDM - analyse seasonal or pricing-related sales.
Owner adjust the price or improve its qualinty.

iii) James R. Evans, 2021

~~5~~

5) <u>Big Data</u>

Large and complex datasets which are not effictively analysed using traditional tools like excel spreadsheets

(ii) Example - Online transactions from FNB is collected and turned into Big data. They use the information to ~~deet~~ detect fraud by analysing spending patterns in real time.

iii) Longbing Cao, 2016

6) <u>Descriptive Analytics</u>

Refers to data analytics that uses historical statistics to describe the data used to gain information for useful purposes

ii) Example - Shoprite checks a week's sale to find the best-selling product of the week.

iii) James R. Evans, 2021

Powered by CamScanner

7) <u>Diagnostic Analytics</u>
Is proces analysing data to underotand why patterns, trends and relationships in data happened.
(ii) Example - At FNB many customers stopped using mobile app. Diagnostic analysis -> finds out app is not user friendly, negative customer reviews. Concludes - complex user interface caused drop in usage.
iii Thomas H. Davenport, 2006

8) <u>Predictive Analytics</u>
Using data analysis tools to predict unkown future events and giving reasons for the prediction.
iii) Example - Shoprite uses past sales data to predict which products will sell more during festive season or in the middle of the month. Therefore better stock planning and reducing shortages or waste.
iii) Dr. S. Bharathi et al, 2026

9) <u>Presciptive Analytics</u>
Optimising data, predictions and indications for recommend actions for smart decision making
(ii) Example: Uber predicts high demand in an area then adjusts prices / ride fares accordingly.
iii) Longbing Cao, 2016

10) <u>Data Literacy</u>
Understanding what data is telling me.
(ii) Example: Business owner sees revenue increased by 30% but profits remained the same. Data-Literate person check costs, discounts, waste to understand why profits didn't increase.
iii) James R. Evans, 2021

Section 2 Data, Measurement and Reporting

11) Structured Data - data organised into fixed data formats in tables, rows, columns. Example - spreadsheets

Student ID	Marks	Name	Subject
001	55%	John	Maths
002	85%	Peter	English

iii) (IBM, 2025)

12) Unstructured Data - data that doesn't have a fixed format, is messy and comes in different forms.
It is hard to analyse.
(ii) Example - Social media images, videos and comments.
iii) (IBM, 2025)

13) Data warehouse - A large central system for storing datasets, databases from different sources. The data is used for BI and data analysis to make complex decissions.
(ii) Example - Shoprite collects data from online transactions, stock levels, customer loyalty programs, gift cards. All data is sent to warehouse, where its cleaned and process
iii) Ian H. written et al, 2011

14) Data Mining - the process of discovering patterns in data. the process is automatic and the patterns discovered are meaninful.
(ii) Example - FNB uses data mining to identify customers likely to miss loan payments.
iii) Ian H. written et al, 2011

15) Data Visualisation - the practice of translating data into visual formats such as charts, graphs, dashboards to simplify complex data. Turns numbers into pictures

ii) Example - An analyst using a bar graph to compare sales across ~~woolworths~~ woolworths clothing stores in Gauteng.
iii) Dr. S. Bharadhi, 2026

16) Dashboard - A visual display showing charts, key performance indactors (KPIs) and matrics in on place. It is presented to executives for them to make business decisions.
(ii) Example - Shoprite manager dashboard showing top-selling products, sale pick hours, daily sales per store and customer reviews.
iii) Dr. S. Bharadhi, 2026

17) Metric - measurement or numbers used to review systems or ~~pe~~ business performance.
(ii) Example - Maize farmer using metrics like number of seed planted, bags of fertilizers used and number of corns haverested. This shows haverst yields.
iii) David Parameter, 2010

18) Key Performance Indicator (KPIs) - are sets of measures focusing on ~~how~~ aspects of organisational performance that are most critical for its current or future success. Used to track success.
(ii) Example - Retail shop tracking Daily sales revenue, profit, customer loyalty and traffic,
KPIs are most important matrics (number) used to track business sucess / goals. Matrics are any measurable numbers.
iii) David Parmeter, 2010

19) Benchmark - Comparing company performance, structure to the best competitors in the industry
ii) FNB comparing mobile app performace ~~against~~ against ~~Ab~~ Absa.

iii) (Investopedia, 2024)

20) <u>Segmentation</u> - Grouping, customers, clients, data, markets into smaller groups based on shared characteristics.
(ii) Example - Shoprite Group segments customers based on income level. Open usave in low income areas, Checkers in middle to high income areas.
iii) (IBM, 2025)

<u>Section 3 : Customer and Sales KPIs</u>

21) <u>Conversion Rate</u> - Is a KPI to measure many customers

$$\text{conversion rate} = \frac{\text{number of conversions}}{\text{total visitors or leads}} \times 100$$

~~turns~~ completes a desired business goal.
ii) Example - Online shop has 5 customers with items in the shopping cart. 3 customers purchase. Conversation rate = 60%
iii) (HubSpot, 2024)

22) <u>Customer Aquisition Cost (CAC)</u> - total cost a business spends to gain a new customer.
(ii) Example: Ligit spends R5000 on ads, gains 10 new customers
$CAC = 5000/10 = R500$ per customer.
iii) David Parmeter, 2010

23) <u>Customer lifetime value (CLV/LTV)</u> - total profit gained per customer over the time they stay a customer.
(ii) Example: A customer buys groceries monthly from Pick n pay

over 20 years. Groceries bought per month are worth R1000
CLV = 1000 × 12 × 20 = R240 000

CLV : CAC ratio = $\frac{\text{CLV} = (\text{how much money customer brings over time})}{\text{CAC} = (\text{cost of acquring a customer})}$

if ratio is significantly large like 20:1 , business has achieved its goal.

iii) (Castéran et al, 2017)

24) Churn Rate - the percentage of customers who leave or stop using a product over time.

ii) Example : Mrp starts the year with 20 000 customers, 100 customers leave. Churn rate = 100 / 20000 × 100 = 0,5%

iii) (Rodriguez-Pérez et al., 2020)

25) Customer Rention Rate (CRR) : % of customers a business keeps over specific time period.

ii) Example : FNB starts 2025 with 1000 customers gains 1000 by 2026 ~~Re~~ ~~End~~ Beginning of 2026 they have 2000 customers.
Customer Rention Rate = $\frac{2000 - 1000}{100}$ = 100% no customer lost.

iii) (Castéran et al., 2017)

26) Net Promoter Score (NPS) : A score used to measure customer satisfaction loyalty and how likely they are to recommend you.

ii) Example : Mac Donalds has a review and online rating, they asks how likely you would recommend them to others.

iii) (Rodriguez-Perez et al.; 2020)

27) Average Order Value (AOV) - Average amount a customer spends per transaction / Revenue per transaction
AOV = total Revenue / Number of orders.

ii) Example : customer buys 10 items, total cost = R2000
AOV = R2000/10 = R200

iii) (Rodriguez - Pérez et al, 2020)

28) Sales Funnel : is a model showing time taken from a customer being aware of the business to the time of making a purchase.

ii) Example : Mbali sees Brightlearn facebook advert, visits website (interest), checks prices and reviews. Subscribes after a week (conversion). subscribes again next month (retention).

iii) (Castéran et al., 2017)

Section 4 Marketing and Campaign Analytics

29) Marketing Campaign - actions / activities taken by a business to promote their product, service to a specific group of people over some period of time.

ii) Example - Shoprite runs a weekend special campaign, send promotion sms to customers, post ads on socials.

iii) (IBM, 2025)

30) Post - Campaign Analysis : is when a business checks its sales, conversion rate and costs after a campaign to see if the campaign wass a success.

(ii) Example - Pick n pay month end specials to increase sales. After campaign they analys the customer traffic, revenue, brand visibility and AOV. and CAC to CLV ratio.
iii) (HubSpot, 2024)

31) <u>Incremental Revenue</u> - Is the extra revenue a company/business makes caused by a promotion or marketing campaign.
i) Example - If pick n pay normally makes R10000 sales from cooking oil after a promotion/campaign they make R12 000 sales in cooking oil. Incremental revenue from campaign = R2000
iii) (HubSpot, 2024)

32) <u>Return on Investment (ROI)</u>: how much profit a business earned compared to how much they spend.
i) Example - A business spends R500 in adverts and campaigns and they make R900, their profit is R400. ROI gained = R400
iii) (Investopedia, 2024)

33) <u>Return on Ad Spend (ROAS)</u>: how much ~~profit~~ Revenue a business earns from the money it spends on adverts.
i) Example: A company spends R60 on ads and makes R200 in sales, the ads brought 3,2 x more the money spend.
iii) (IBM, 2025)

34) <u>Click-Through Rate (CTR)</u> - is the percentage of viewers who click on an ad or link after seeing it.
i) Example - If 1000 people sees an add 100 click on it, the ~~CR~~ CTR = 10%

iii) (Investopedia, 2024)

35) <u>Bounce Rate</u> - % of viewers or visitors who visits a website and leave without interacting or clicking anything.
ii) Example - R10 people visit a page and 4 leave without clicking anything. Bounce rate is 40%.
iii) (Investopedia, 2024)

36) <u>A/B Testing</u> - is when a business compares two ads, websites to see which one performs better.
ii) Example - company shows version A to group 1 of people and version B to group 2 of people. Then checks which one gets more clicks and sales.
iii) (HubSpot, 2024)

37) <u>Lead Generation</u> : Is getting potential customers to show interest in a business by attracting their interest in a product or service.
ii) Example : A business gets people to sign up with their email so they can later send adverts or marketing campai[gn] posts to them so as to sell something.
iii) (HubSpot, 2024)

38) <u>Marketing Attribution</u> - tells which marketing activity (email, socials campaigns) led the customer to buy or make the business' desired goal.
ii) Example : Business makes flyers, and social posts. Customer buys product after seeing a flyer: The flyer gets credit.
iii) (Caste'ran et al., 2017)

ii

References

1) Longbing Cao (2016) 'Data Science: A comprehensive Overview', ACM Computing Surveys 1,1(1), pp. 3-4
2) Thoms H. Davenport (2006) 'Competing on Analytics', Havard Business Review
3) James R. Evans (2021), Business Analytics Methods, Models, and Decisions. 3rd edn. University of Cincinnati. Pearson
4) Ian H. Written, Frank, E. and Hall, M, A. (2011) Data Mining Pratical Machine Learning Tools and Techniques. 3rd edn. United States. Elsevier Inc
5) David Parmenter (2010), Key Performance Indicators Developing, Implementing, and Using Winning KPIs. 2nd edn. Canada. John Wiley and Sons, Inc.
6) Dr. S. Bharathi, Dr. Dhaneesh. V (2020) Advanced Business Analytics: Models, Methods and Applications. 1st edn. Dhanalakshmi Srinivasan University. Bhumi Publishing.
7) IBM (2025) Structured vs. unstructured data: https://www.ibm.com/think/topics/structured-vs-unstructured-data. (Accessed: 27 April 2026).
8) Investopedia (2024) Benchmark, Available at: https://www.investopedia.com (Accessed: 27 April 2026).
9) IBM (2025) What is customer segmentation? Available at: https://www.ibm.com (Accessed: 27 April 2026)
10) HubSpot (2024) Conversion Rate. Available at: https://www.hubspot.com (Accessed 27 April 2026)
11) Castéran, H., Meyer-Waarden, L. and Reinartz, W. (2017) Modeling Customer Lifetime Value, Retention, and Churn. Available at https://www.researchgate.net/publication/318167419 (Accessed: 27 April 2026)
12) Rodriguez-Pérez, G. et al (2020) 'Churn and Net Promoter Score forecasting for business decision-making', Knowledge-Based Systems.